

Approach to a Patient Coming with Acute Chest Pain

Section: Aortic Dissection · History after Stabilisation · Other Differential Diagnoses of Chest Pain

1. Aortic Dissection

1.1 Definition & Classification

Acute aortic dissection is defined as a tear in the aortic intima allowing blood to enter between the layers of the aortic wall, creating a false lumen and leading to propagation of the dissection. ([NCBI](#))

The most common classification schema is the Stanford system:

- *Type A* involves the ascending aorta (proximal to the brachiocephalic branch) — requires emergent surgical management. ([NCBI](#))
- *Type B* involves the descending aorta (distal to the left subclavian artery) — often managed medically if uncomplicated. ([StatPearls](#))

1.2 Why It Matters

This is a life-threatening vascular emergency. Without treatment, the mortality is very high (≈ 50% within 48 h for ascending dissections). ([Medscape](#))

Rapid recognition, diagnosis, and appropriate intervention (surgical or medical) are essential for survival. ([Ahajournals](#))

1.3 Clinical Presentation

Pain characteristics: Often sudden onset, severe, described as “tearing,” “ripping” or “sharp,” located in the chest, back or between the shoulder blades, sometimes migrating as the dissection extends. ([PubMed Central](#))

Associated symptoms: Syncope, acute neurologic deficits (stroke-like), limb ischemia, visceral/renal ischemia, acute heart failure (if aortic regurgitation or tamponade), new murmur of aortic regurgitation. ([NCBI](#))